

Graphplay

A Machine-Checked Theory of Equitable-Partition Quantum Walks, with Verified Application to Attention Complexity and Hardware

Graphplayers Crew — Graphplay v3

Abstract

We present a Lean 4 formalization of the theory of continuous-time quantum walks (CTQW) on graphs carrying an **equitable partition**, and three verified consequences that, to our knowledge, have not appeared in machine-checked form before. The mathematical spine — a CTQW restricts to the cell-constant subspace and acts there as the walk on a small symmetric quotient $\tilde{Q} = D^{1/2}QD^{-1/2}$, lifting perfect state transfer (PST), spectrum, search, and mixing — is classical (Godsil; Bachman–Tamon et al.; Ide–Narimatsu). Our contribution is a **certificate**: the spine and its consequences are sorry-free Lean theorems with an explicit proof ledger.

On top of the spine we prove three results that are genuinely ours. First, an **exact** $O(n^2) \rightarrow O(n \cdot r)$ reduction of the attention operation under an equitable token pattern, machine-checked for the forward pass, the backward pass, and a full training step. Second, a **verified compiler**: a typed transformer DSL whose denotation commutes with compilation to the quotient host (`compile_denote_commutes`, `compiler_guarantee`, both sorry-free) — the functor stack **is** the compiler, and its correctness is a theorem. Third, an **exact finite- n** quantum-search advantage — amplitude $\sqrt{(n-1)/n}$ on the complete graph in time $O(\sqrt{n})$ against a matching classical lower bound — lifted to structured machine-learning search, together with a verified advantage on the **sparse, buildable** hypercube and an honest statement of the $d > 4$ lattice threshold.

We then disassemble two real devices in Lean: IBM’s heavy-hex lattice (a 2×2 data/flag quotient with eigenvalues $\pm 2\sqrt{N-1}$ and three engineering payoffs) and Microsoft’s Majorana 1 (a parity-sector quantum equitable partition). Finally we propose — as the paper’s last and most speculative move — a falsifiable on-device experiment that the compiler emits for IBM Heron. We are scrupulous throughout about the boundary between what is proven and what is conjectured; the open frontier (Conjecture 9.3, the relativistic enrichment, the ∞ -categorical towers) is flagged as open.

1. Introduction

The contribution. This paper is a machine-checked theory of equitable-partition quantum walks, together with its verified application to two things people actually care about: the cost of attention in transformers, and the structure of real quantum hardware. The whole development lives in Lean 4; the load-bearing claims are sorry-free theorems, and we say explicitly, theorem by theorem, which is which.

The organizing observation is old and robust: a continuous-time quantum walk on a graph G that carries an **equitable partition** π — a coloring whose cells have constant inter-cell degrees — never leaves the cell-constant subspace if it starts there, and on that subspace it evolves as the walk on a small **quotient** operator. Two decades of the Tamon program, and Godsil before it, show that PST, spatial search, and mixing on large graphs are very often shadows of a tiny finite calculation seen through this lens. We take that body of mathematics as given and known (Section 6 attributes it carefully). What we add is a **certificate layer**: a uniform, axiom-clean Lean spine (Section 2) and three verified consequences (Section 3) that the pencil-and-paper and empirical literature states without proof or without exactness.

Why a certificate. The headline applications — “structured attention costs $O(nr)$, not $O(n^2)$ ”, “this ML program compiles to that quantum walk with the same semantics”, “this search reaches the marked state in $O(\sqrt{n})$ with amplitude exactly $\sqrt{(n-1)/n}$ ” — are the kind of claim that is easy to assert and hard to get exactly right. By making each one the statement of a Lean theorem that take build checks, we turn the assertions into artifacts. The honest cost is that some deep **rate** clauses (the spectral-gap timing of search on sparse hosts, the $d > 4$ lattice convergence) remain honest sorrys; we mark them as such and never lean on them.

Structure. Section 2 states the verified spine and its lift theorems. Section 3 gives the three novel verified results: the exact attention reduction, the verified ML-to-host compiler, and the exact finite- n search advantage plus its sparse-host versions. Section 4 disassembles two real chips. Section 5 proposes the on-device experiment (deliberately last; it is the one part that is a proposal, not a proof). Section 6 is the honest related-work and novelty accounting: we are a **certificate, not a first contact**. Section 7 is the open frontier.

Intended frontend. The verified backend described here is meant to sit under a practical deep-learning compiler. Our target frontend is **catgrad** (hellas.ai), whose open-hypergraph categorical SSA IR is a natural source language: a Graphplay backend lowers catgrad’s categorical IR to the host representation of Section 3, so that the equitable-reduction and hardware-compilation theorems apply to programs a practitioner actually writes. The integration is design-stage; see the forward reference paper/catgrad_integration.md (planned).

2. The Verified Spine

2.1 Equitable partitions and the symmetric quotient. A weighted graph on a finite vertex set V is a Hermitian $A \in \mathbb{C}^{V \times V}$. A partition π of V into nonempty cells $C_1, \dots, C_r$ is **equitable** for A when, for every ordered pair of cells (C_j, C_k) , the row sum $\sum_{w \in C_k} A_{vw}$ is independent of the choice of $v \in C_j$. Writing $D = \text{diag}(|C_1|, \dots, |C_r|)$ and Q for the raw quotient of inter-cell row sums, the object we actually walk on is the **symmetric (normalized) quotient**

$$\tilde{Q} := D^{1/2} Q D^{-1/2},$$

which is Hermitian even though Q need not be. This normalization is the technical pivot of the whole development — it is what makes the lifted operator self-adjoint, hence a legitimate Hamiltonian, in every tower. Lean carriers: Graphplay/Equitable.lean (symmQuotient, symmQuotient_isHermitian), Graphplay/Weighted.lean.

2.2 The restriction theorem. The keystone is that the walk restricted to the cell-uniform subspace is the walk on $\tilde{Q}$. Concretely, for a cell-uniform vector $\sum_i w_i u_i$ (with u_i the rescaled cell indicators),

$$A \cdot \left(\sum_i w_i u_i \right) = \sum_i (\tilde{Q} w)_i u_i.$$

This is EquitablePartition.restrict_eq_symmQuotient (**PROVEN**, sorry-free, Graphplay/Equitable.lean). From it the rest of the spine follows.

2.3 The lift theorems. Each is an axiom-clean Lean theorem.

Lift	Statement	Lean (PROVEN)
Spectrum	$\text{spec}(\tilde{Q}) \subseteq \text{spec}(A)$; quotient eigenpairs inflate to cell-constant eigenpairs of A	spectrum_subset (Spectral.lean)

PST (iff)	cell-uniform PST on (A, π) holds at time τ iff $\tilde{Q}$ has PST at τ — both directions	cellUniformPST_iff_quotientPST (PST/QuotientIff.lean)
Search	the marked-cell-refined search Hamiltonian restricts to the $(r\{+\}1)$ -dim cell-uniform-plus-marked subspace	search_quotient_reduction (Search.lean)
Products	Cartesian-product PST factors through the Kronecker-sum exponential	cartesianProduct_pst (Product/PST.lean)

The PST iff is the Lean form of Bachman–Tamon et al. (arXiv:1108.0339): a graph has cell-uniform PST exactly when its symmetric quotient does. We prove both directions sorry-free. The spectrum lift is Godsil’s classical inclusion. Together these are the instruction set the rest of the paper compiles to.

2.4 Symmetry produces equitability. A graph automorphism induces an equitable partition by its orbits: equitableOfAutomorphism (Graphplay/Integrations/MachineLearning.lean, **PROVEN**). This is the bridge that lets a **symmetry** in a model (weight-tying, permutation-equivariance) become a **quotient** we can reduce against — the hook Section 3 uses for attention.

3. Novel Verified Results

This section contains the results that are ours. Each is a sorry-free Lean theorem; the few deep rate clauses we depend on nowhere are flagged where they appear.

3.1 Exact $O(n^2) \rightarrow O(n \cdot r)$ attention reduction. Cast single-head attention as application of a token-token score operator on n tokens of width d . If the score pattern is **equitable** with r cells (weight-tying, block structure, or an equivariance orbit via §2.4), the $O(n^2d)$ full application equals a **block** application that costs $O(nrd + nd)$.

The forward identity is exact, not approximate:

$$\text{blockAttentionApply} = \text{fullAttentionApply}$$

(blockAttentionApply_eq_fullAttentionApply, **PROVEN**, sorry-free), with cost blockCost = $nrd + nd$ linear in n (attention_apply_linear_in_n, **PROVEN**) and blockCost_le_fullCost for $r \leq n$ (**PROVEN**).

The reduction is verified through **training**, not just inference:

- **Backward pass.** The block gradient computes the same gradient as the dense backward (blockGrad_apply, **PROVEN**).
- **Training step.** One full step (forward + backward + update) is linear in n under a maintained equitable structure (training_step_linear_under_equitable, **PROVEN**).

Lean: Graphplay/Integrations/AttentionComplexity.lean. The structural identity is exact and sorry-free; we make **no** claim about a deeper “quantum-rate” composition here (the speculative attention_quantum_composition clause carries an honest sorry and is not used).

3.2 The functor stack is a verified compiler. We define a small typed transformer DSL (TransformerProgram) with a denotational semantics denote and a compilation pass compile that targets the symmetric quotient host. The compiler is correct by theorem.

- compile_denote_commutates (**PROVEN**, **zero sorries**): the diagram commutes — compiling then denoting equals denoting then evaluating on $\tilde{Q}$.

- `compiler_guarantee` (**PROVEN**, **zero sorries**): the compiled program is **both** semantics-preserving **and** complexity-reducing (linear in n).

This is the wedge against the empirical “quantum-walk transformer” literature (Section 6): there, the walk is a heuristic feature; here, the compiler from ML semantics to walk host carries a machine-checked commuting-semantics proof. Lean: `Graphplay/Integrations/TransformerDSL.lean`. The multi-head pooled head restricts to the symmetric quotient (`multiHead_restrict_eq_symmQuotient`, **PROVEN**, `Integrations/MachineLearning.lean`). A concrete ML-to-device specialization (`compiled_cellUniform_realizes_target`, `compile_recall_to_heron`, **PROVEN**) lives in `Graphplay/Applications/CompileML.lean` and feeds Section 5.

3.3 Exact finite- n quantum search advantage. On the complete graph K_n with one marked vertex, the search dynamics live in an exact 2-dimensional invariant subspace (`completeGraph_2d_block`, **PROVEN**). We extract the **exact** finite- n amplitude:

At evolution time $t_q \leq (\pi/2)\sqrt{n}$, the exact search amplitude is at least $\sqrt{(n-1)/n} \geq \sqrt{1/2}$ (`quantum_search_exact_amplitude`, **PROVEN**, sorry-free), against the classical fact that fewer than n queries cannot certify the marked element (`classical_search_lower_bound`). The pair is `quantum_search_quadratic_advantage` (**PROVEN**).

Lifting along §2.4: a structured ML search over r configurations inherits the exact advantage — amplitude $\geq \sqrt{1/2}$ in time $\leq (\pi/2)\sqrt{r}$ versus an r -query classical bound (`ml_structured_search_quantum_advantage_exact`, **PROVEN**). The matching quantum **optimality** (the $\Omega(\sqrt{N})$ side) is Bennett–Bernstein–Brassard–Vazirani; we cite it and do not reprove it. Lean: `Graphplay/Integrations/QuantumAdvantage.lean`.

3.4 Advantage on sparse, buildable hosts. The complete graph is not buildable. The point of this subsection is that the **same** equitable-quotient route gives the advantage on hosts with bounded degree.

- **Hypercube (proven half).** The Boolean hypercube Q_d is d -regular with $d = \log_2 N - \log\text{-degree}$, hence physically realizable — and its Hamming-distance partition into $d + 1$ cells is equitable; the marked search Hamiltonian collapses to the $(d\{+\}1)$ -dimensional binomial chain. Both the **spar-sity** and the **exact equitable reduction** are sorry-free (`hypercube_sparse_search_advantage`, the regularity and quotient-reduction conjuncts). The deep $O(\sqrt{N})$ **timing** clause leans on the Childs–Goldstone / CNO spectral-ratio analysis (arXiv:2004.12686) and is the **single honest sorry** in this theorem — we mark it **OPEN** and do not use it downstream.
- **Lattice threshold (honest).** The d -dimensional lattice is regular and sparse (`latticeGraph_isRegular`, `latticeGraph_sparse`, **PROVEN**). The statement `lattice_search_dimension_threshold` — that the advantage appears for $d > 4$ and fails below — is a genuine formalized statement; its spectral half is an honest sorry (**OPEN**), while the low-dimensional **no-advantage** contrast (`buildable_lattice_no_advantage_low_dim`) is structurally in place. This is the honest Childs–Goldstone $d = 4$ threshold, not a claimed proof of it.

Lean: `Graphplay/Applications/SparseSearch.lean`. The honesty here is itself the point: the **structural** reductions are verified; the **dynamical timing** of search is exactly where the open mathematics lives, and we put the sorry there in plain sight.

4. Applications: Verified Spectral Disassembly of Real Chips

The spine runs **backward** too: given a real device Hamiltonian, recover its equitable structure and read off the small quotient. We do this for two chips, in Lean.

4.1 *IBM heavy-hex*. The heavy-hex lattice admits a data/flag equitable partition. Its symmetric quotient is a 2×2 Hermitian matrix whose spectrum we compute exactly:

$$\text{spec}(\tilde{Q}_{\text{data/flag}}) = \{\pm 2\sqrt{N-1}\}, \quad N = |\text{HoneyVertex}| = 2nm$$

(dataFlagQuotient_eigenvalues, **PROVEN**, sorry-free).

The idealized toroidal **template** has the toy value $\pm\sqrt{6}$ (the weighted K_2 with edge $\sqrt{6}$); the concrete site-subdivided chip corrects this to the proven $\pm 2\sqrt{N-1}$. Three engineering payoffs follow on the quotient (Graphplay/Applications/IBMHeavyHex.lean):

1. **PST** between the data cell and the flag cell, via heavyHex_pst_lift (**PROVEN**), at the quotient PST time;
2. **uniform mixing** on the quotient at the corresponding mixing time;
3. **dephasing robustness**: a data/flag-symmetric dephasing channel preserves the partition (dephasing_preserves_dataFlag, **PROVEN**), and the real (non-chiral) data/flag quotient gives **no** search speedup (dataFlag_chiral_no_speedup, **PROVEN**) — an honest negative result.

4.2 *Majorana 1*. The Microsoft Majorana 1 fabric is parity-conserving. We build the parity-sector projectors and prove they resolve the identity (sectorProjector_sum, **PROVEN**), assemble them into a **quantum** equitable partition (parityQuantumEquitablePartition, **PROVEN**), and show a parity-respecting noise model preserves it (payoff2_parity_noise_preserves_partition, **PROVEN**). This is the Tower-3 (operator-algebra) face of the spine: the partition is by a *-subalgebra of parity sectors, not by vertex cells. Lean: Graphplay/Applications/MajoranaOne.lean. We flag the physical caveat (Majorana zero modes are, on current superconducting hardware, **simulated** rather than natively realized; cf. Nature Commun. 2023) and treat this disassembly as a verified **modeling** exercise.

5. Proposed Experiment (a Proposal, Not a Proof)

Everything above is proven. This section is the one part that is a **proposal**: a falsifiable experiment the compiler emits for existing hardware. We place it last on purpose.

The CompileML protocol on IBM Heron. Compile a key-value recall task to a heavy-hex CTQW host (compile_recall_to_heron, **PROVEN**, as a Lean object). The on-device protocol is:

1. **Prepare** a query-uniform initial state (uniform over the data cell);
2. **Evolve** under the native heavy-hex CTQW for time $\pi/(2q)$, where q is the quotient coupling;
3. **Measure** the key-uniform (flag-cell) population.

The compiler's prediction (compiled_experiment_prediction, **PROVEN** as the structural identity) is that the measured key population equals the squared quotient amplitude. Crucially, the protocol is **falsifiable**: if the device's true coupling pattern breaks the data/flag partition, a nonzero **BreakingScore** appears, and the predicted population deficit is proportional to it (compiled_breakingScore_zero_blockDiagonal, and the falsification witness compiled_positive_breakingScore_exists). The experiment thus tests the partition hypothesis on the real chip rather than assuming it. Lean: Graphplay/Applications/CompileML.lean. The intended path from a practitioner program to this protocol runs through the catgrad frontend (paper/catgrad_integration.md, planned).

6. Related Work and Honest Novelty

We are a **certificate, not a first contact**. The mathematics of our spine, and of every physical phenomenon we lift, is prior art; our wedge is the machine-checked exactness, the verified compiler, and the falsifiable experiment. We attribute carefully.

The spine. The equitable-partition-to-quotient lift, and PST-iff-quotient in particular, is **Bachman–Fredette–Fuller–Landry–Opperman–Tamon–Tollefson** (arXiv:1108.0339) on a base of **Godsil’s** spectral and PST theory (arXiv:1102.4898). The unification of equitable partition, PST, and CTQW search is **Ide–Narimatsu** (arXiv:2209.07688) — the single closest prior work to our thesis, and the source of the mathematical content we formalize. We claim none of this mathematics; we claim its Lean certificate.

Sparse-host search. CTQW spatial search on K_n and the hypercube is **Childs–Goldstone** (quant-ph/0306054). That a **local** equitable symmetry (not a global automorphism) suffices for fast search is **Janmark–Meyer–Wong** (arXiv:1403.2228) — exactly our “equitable, not automorphism” framing. The success-probability/finding-time reduction via equitable partition is again **Ide–Narimatsu** (arXiv:2209.07688). The $\Omega(\sqrt{N})$ optimality is Bennett–Bernstein–Brassard–Vazirani. Our hypercube_sparse_search_advantage and the lattice threshold formalize this corpus; the timing sorrys sit precisely on the analytic content (arXiv:2004.12686) that these works supply on paper.

Attention as a quantum walk. Marrying a quantum walk to transformer attention is already done: **GQWformer** (arXiv:2412.02285) and **CTQWformer** (arXiv:2605.09486). Both are **classical and empirical** — the walk is an inductive bias or a precomputed feature; neither claims, let alone proves, an exact complexity reduction, and both retain $O(n^2)$ attention. The bare bridge is theirs; we cite it as motivation, not competition.

Our wedge, in one sentence. We are the first to (a) cast structured attention itself as the equitable token-graph operator and **machine-check** its exact $O(n^2) \rightarrow O(nr)$ reduction across forward, backward, and training step; (b) expose that as a **typed ML-to-host compiler with a proven commuting semantics**; and (c) emit from the same compiler a **falsifiable on-device PST experiment**. The honesty about which clauses are sorry-ed (the deep search **rates**) is itself part of the contribution: it is what distinguishes a certificate from a slogan.

7. Open Frontier

We are explicit that the following are *not* proven. They are the program’s horizon, and we state them as open.

7.1 Conjecture 9.3 (Bose–Mesner $\equiv$ chiral $\equiv$ graphon). A consistent family (G_n, π_n) admits **both** a graphon quasi-infinite limit **and** a strict chiral-signing mixing speedup *iff* its Bose–Mesner algebra eventually coincides with the partition-projector algebra $\mathcal{P}(\pi_n)$. The Lean apparatus (Graphplay/Dowsing/Conjecture93.lean) builds **six genuine test families** as real WeightedGraph objects with honestly-proven equitable partitions — K_n^σ , the Heawood envelope, Hamming schemes, $K_{n,n,n,n}$, $K_n + P_n$, and Cayley(S_n) by transpositions. But **both halves of the conjecture itself are sorry** (Conjecture93_weak and its forward/reverse directions); the only family verdict proven outright is the both-sides-false Cayley(S_n) case (False $\leftrightarrow$ False). **OPEN** We claim the **families and the statement**, not the theorem.

7.2 Relativistic / Dirac enrichment. A Dirac-operator (first-order, spinorial) enrichment of the spine, with a continuum limit, is sketched but the continuum limit is an honest sorry. **OPEN**

7.3 The ∞ -categorical towers. Towers 6–7 (sheaves of weighted graphs; stable- ∞ -categorical Hermitian endomorphisms with coherent idempotent partitions) are **definitional scaffolds only**. The key

colimit-preservation obligation (`Quotient.mapCocone_isColimit`) is an irreducible sorry blocked on missing Mathlib ∞ -categorical infrastructure. These towers are **not proven** and we make no theorem-level claim about them. **OPEN**

7.4 Other deferred analytic content. We record, as open, the graphon self-adjoint spectral-measure gaps (some Tower-4 spectral statements carry sorries on the measure side), the Tsirelson bound (the quantum CHSH value; note the **classical** CHSH value $3/4$ is proven, `CHSH_classical_value`, in `Graphplay/QuantumCSP.lean`), and the Lovász- θ SDP-duality direction.

7.5 Toward the practical frontend. The intended path to use is a Graphplay backend under **catgrad** (hellas.ai): lower its open-hypergraph categorical SSA IR to the host representation of Section 3 so that the verified attention reduction and the hardware compiler apply to real models. This is design-stage (paper/`catgrad_integration.md`, planned), and the quantitative noisy-CTQW and Lindbladian extensions of the spine remain open beyond the cell-uniform invariance already proven.

A closing note. The pleasure of this project was watching analytic miracles — transfer at an integer time, search that finds in $\sqrt{N}$ steps — dissolve into small finite calculations seen through the right partition, and then watching Lean agree. What we offer is not a new miracle but a ledger: here is the structure, here is the proof, and here, marked in the open, is the part we have not yet earned.

References

Spine — equitable partitions, PST, quotients:

- C. Godsil. State transfer on graphs, arXiv:1102.4898.
- R. Bachman, E. Fredette, J. Fuller, M. Landry, M. Opperman, C. Tamon, A. Tollefson. Perfect state transfer on quotient graphs, arXiv:1108.0339 (QIC 12, 293).
- Y. Ide, A. Narimatsu. Perfect state transfer, equitable partition and continuous-time quantum walk based search, arXiv:2209.07688.

Search — sparse hosts, optimality, timing:

- A. Childs, J. Goldstone. Spatial search by quantum walk, quant-ph/0306054 (PRA 70, 022314).
- I. Janmark, D. Meyer, T. Wong. Global symmetry is unnecessary for fast quantum search, arXiv:1403.2228 (PRL 112, 210502).
- S. Chakraborty, L. Novo, J. Roland. On the optimality of spatial search by continuous-time quantum walk, arXiv:2004.12686 (PRA 102, 032214).
- C. Bennett, E. Bernstein, G. Brassard, U. Vazirani. *Strengths and weaknesses of quantum computing*, SIAM J. Comput. 26 (1997) (BBBV lower bound).

Attention as quantum walk (classical / empirical, concurrent):

- Yu, Chen, Lv, Yang (Zhejiang Lab). GQWformer: A quantum-based transformer for graph representation learning, arXiv:2412.02285.
- CTQWformer: A CTQW-based transformer for graph classification, arXiv:2605.09486.
- S. Aaronson. Quantum machine learning: read the fine print, Nature Physics 11 (2015).

Hardware:

- IBM Quantum. The IBM Quantum heavy-hex lattice, 2021.
- Observing and braiding topological Majorana modes on a superconducting processor, Nature Commun. 14 (2023).
- Perfect state transfer on superconducting transmons with tunable couplers, Nature Commun. 16 (2025).

Frontend (intended):

- catgrad, hellas.ai — open-hypergraph categorical deep-learning compiler. Graphplay-backend integration: `paper/catgrad_integration.md` (planned).

Companion material:

- Graphplayers Crew. *Graphplay research program* (`paper/research_program.typ`): the dowsing-rod open theorems, cross-disciplinary integrations, conjectural Towers 6–7, and the full Conjecture 9.3 analysis.